

Adopted Levels, Gammas

$Q(\beta^-)=14170\ 40$; $S(n)=5297\ 8$; $S(p)=15831\ 9$; $Q(\alpha)=-14894\ 8$ [2021Wa16](#)

$S(p), Q(\alpha)$: Deduced by evaluators using mass excesses of $8318\ 5$ for ^{34}Mg : a weighted average of $8323\ 7$ ([2019As04](#)) and $8315\ 5$ ([2025Ly01](#)), and $12245.3\ 26$ for ^{31}Na measured by [2025Ly01](#); $-224\ 7$ for ^{35}Al from [2021Wa16](#). Values from [2021Wa16](#): $S(p)=15836\ 10$, $Q(\alpha)=-14895\ 16$.

$S(2p)=38400.5\ 85$, $Q(\epsilon)=-15753\ 10$, from mass excesses of $23598.8\ 43$ for ^{33}Na and $15529.5\ 71$ for ^{35}Mg measured by [2025Ly01](#); $-224\ 7$ for ^{35}Al from [2021Wa16](#). Values from [2021Wa16](#): $S(2p)=38580\ 450$, $Q(\epsilon)=-15860\ 270$.

$S(2n)=7869\ 10$, $Q(\beta^-n)=11697\ 7$ ([2021Wa16](#)).

Isotope discovery ([2012Th10](#)): $\text{C}(^{40}\text{Ar}, \text{X})$ projectile fragmentation at Berkeley ([1979Sy01](#)).

^{35}Al production:

[2015Mo17](#): $^9\text{Be}(^{40}\text{Ar}, \text{X})$ at $E(^{40}\text{Ar})=95$ MeV/nucleon at RIKEN. Measured angular distributions and transverse momentum distributions of fragments. Deduced formulation for the width of transverse momentum distribution as a function of fragment velocity.

[2012Kw02](#): $E(^{40}\text{Ar})=140$ MeV/nucleon impinged on ^9Be , natural nickel, and ^{181}Ta targets at NSCL. Measured fragmentation cross sections, parallel momentum transfers, and widths. Compared with empirical formula EPAX, and predictions from internuclear cascade and deep inelastic models using Monte Carlo ISABEL-GEMINI and DIT-GEMINI codes.

[2012Zh06](#): $^9\text{Be}(^{40}\text{Ar}, \text{X})$ at $E(^{40}\text{Ar})=57$ MeV/nucleon at HIRFL. Measured momentum distributions and production cross sections of fragments. Observed competition between projectile fragmentation and other mechanisms. Compared with EPAX, abrasion-ablation, and HIPSE models. Studied target dependence of fragment cross sections.

[2007No13](#): $^9\text{Be}(^{40}\text{Ar}, \text{X})$ at $E(^{40}\text{Ar})=100$ MeV/nucleon at RIKEN. Measured fragment momentum distributions and production cross sections.

^{35}Al decay measurements:

[2017Ha23](#): $^9\text{Be}(^{40}\text{Ar}, \text{X})$ at HIRFL. Measured $T_{1/2}$.

[2005Ti11, 2006AnZW](#): $(^{36}\text{S}, \text{X})$ at GANIL. Measured $T_{1/2}$, β^- -delayed γ and neutron spectroscopy.

[2001Nu01, 2002Nu02](#): $\text{U}(p, \text{X})$ at CERN. Measured $T_{1/2}$, β^- -delayed γ and neutron spectroscopy.

[1999YoZW](#): $^9\text{Be}(^{48}\text{Ca}, \text{X})$ and $^{181}\text{Ta}(^{48}\text{Ca}, \text{X})$ at RIKEN. Measured $T_{1/2}$ and $\% \beta^-n$.

[1995ReZZ, 2008ReZZ](#): $^{232}\text{Th}(p, \text{X})$ at LAMPF. Measured $T_{1/2}$ and $\% \beta^-n$, and average E_n .

[1989Le16, 1989MuZU](#): $^{181}\text{Ta}(^{48}\text{Ca}, \text{X})$ at GANIL. Measured $T_{1/2}$ and $\% \beta^-n$.

[1988Mu08, 1988MuZY, 1988BaYZ](#): $^{181}\text{Ta}(^{86}\text{Kr}, \text{X})$ at GANIL. $T_{1/2}$ and $\% \beta^-n$.

^{35}Al radius measurement:

[2006Kh08](#): ^{35}Al produced by $^{181}\text{Ta}(^{48}\text{Ca}, \text{X})$ fragmentation at $E(^{48}\text{Ca})=60.3$ MeV/nucleon at GANIL. Measured energy-integrated reaction cross sections at 30-65 MeV/nucleon using a silicon telescope as both active target and detector. Deduced reduced strong absorption radii, isospin dependence, and possible halo structure or large deformation.

^{35}Al knockout-reaction measurements:

[2012No05](#): $^{33,34,35,36}\text{Al}$ produced by $\text{Be}(^{48}\text{Ca}, \text{X})$ fragmentation at $E(^{48}\text{Ca})=900$ MeV/nucleon at GSI. Measured $1n$ removal cross sections and longitudinal momentum distributions of the residues. Deduced single-particle occupancies in the ground states of $^{33,34,35}\text{Al}$. $\sigma(^{35}\text{Al} \rightarrow ^{34}\text{Al})=75$ mb 4 and $\sigma(^{36}\text{Al} \rightarrow ^{35}\text{Al})=95$ mb 5 .

[2010Ro23](#): ^{35}Al produced by $^9\text{Be}(^{40}\text{Ar}, \text{X})$ fragmentation at $E(^{40}\text{Ar})=700$ MeV/nucleon at GSI. Measured $1n$ knockout cross sections and longitudinal momentum distributions of the residues. $\sigma(^{35}\text{Al} \rightarrow ^{34}\text{Al})=65$ mb 18 .

^{35}Al in-beam γ spectroscopy:

[2006FuZX](#): ^{35}Al produced by $\text{Be}, \text{C}(^{40}\text{Ar}, \text{X})$ and observed one 760.1-keV $21\ \gamma$ ray from $\text{He}(^{35}\text{Al}, ^{35}\text{Al}\gamma)$ without placing it in the level scheme.

^{35}Al mass measurements: [2017Ga20](#), [2007Ju03](#), [1991Or01](#), [1991Zh24](#), [1987Gi05](#).

Theoretical calculations (binding energies, deformation, quadrupole moments, radii, levels, J , π , mass, $T_{1/2}$, etc): [2016Sa46](#), [2014Ca21](#), [2013Li39](#), [2013Sh05](#), [2011Ki12](#), [2009Yo05](#), [2004Kh16](#), [1994Po05](#).

 ^{35}Al LevelsCross Reference (XREF) Flags

A	$^{36}\text{Mg}\ \beta^-n$ decay (6.9 ms)	D	$\text{Pb}(^{35}\text{Al}, ^{34}\text{Al}n\gamma)$
B	$^9\text{Be}(^{36}\text{Si}, ^{35}\text{Al}\gamma)$	E	Coulomb excitation
C	$\text{He}(^{35}\text{Al}, ^{35}\text{Al}'\gamma)$		

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Adopted Levels, Gammas (continued) ^{35}Al Levels (continued)

$E(\text{level})^\dagger$	J^π	$T_{1/2}$	XREF	Comments
0	$(5/2)^+$	38.1 ms 4	AB DE	$\% \beta^- = 100$; $\% \beta^- n = 37$ 4; $\% \beta^- 2n = ?$ J^π : From shell-model calculations and $L(^{36}\text{Si}, ^{35}\text{Al}) = 2$ from 0^+ . $T_{1/2}$: weighted average of 38.4 ms 3 (2017Ha23, implant- β correlation), 36.8 ms 5 (2005Ti11, 2006AnZW, implant- β correlation), 38.6 ms 4 (2001Nu01, 2002Nu02, three β -counting rates and one γ -counting rate), 30 ms 4 (1995ReZZ, 2008ReZZ, implant- β/n correlation), 170 ms +90-50 (1989Le16, 1989MuZU, β -decay curve), 30 ms 10 (1988DuZT, beam pulsations), and 130 ms +100-50 (1988Mu08, 1988BaYZ, β counting rate). $\% \beta^- n$: Unweighted average of 41 2 (2005Ti11, 2006AnZW), 41 13 (2001Nu01, 2002Nu02), 26 4 (1995ReZZ, 2008ReZZ), and 40 10 (1989Le16, 1989MuZU). Others: 43 9 (1999YoZW, preliminary), 87 +37-25 (1988Mu08, 1988BaYZ) with a non-physical upper bound. Note that the original value reported in 2005Ti11 is 38 2 (2006AnZW report 38 3) and the evaluators have added intensities of γ rays deexciting the 4379 (0.9% 2) and 4255 (2.0% 8) levels in ^{34}Si not included in their reported $\% \beta^- n$ value, as the authors state that these levels are fed by either neutrons with energy below their detection threshold or by weak transitions that cannot be identified in the neutron spectrum. $\% \beta^- 2n$: No evidence was found for any $A=33$ activity and $2n$ emission was thus found negligible (2001Nu01). Reduced strong absorption radius $r_0^2 = 1.188 \text{ fm}^2$ 14 from the energy-integrated σ of $\text{Si}(^{35}\text{Al}, X)$ (2006Kh08). Major configurations for $J^\pi = 5/2^+$ of ^{35}Al g.s. from $\text{Pb}(^{35}\text{Al}, ^{34}\text{Aln}\gamma)$: (g.s., 4^- in ^{34}Al) $\otimes$ $\nu p_{3/2}$, $S=0.36$ 9 (2017Ch36); (46 keV, 1^+ in ^{34}Al) $\otimes$ $\nu d_{3/2}$, $S=1.47$ 22 (2017Ch36); (1.4 MeV, 2^+ in ^{34}Al) $\otimes$ $\nu s_{1/2}$, $S=0.16$ 1 (2021Bh12); (2.5 MeV, 3^- in ^{34}Al) $\otimes$ $\nu p_{3/2}$, $S=1.48$ 18 (2021Bh12).
802 3			AB	
1007 4			B E	XREF: $E(1020)$ $B(E2)^\ddagger = 0.0142$ 52 from Coulomb excitation assuming E2 excitation mode (1999Ib01). Others: $B(E1)^\ddagger \leq 0.00020$ 9, $B(E2)^\ddagger \leq 0.0125$ 56, $B(M1)^\ddagger \leq 0.0024$ 11.
1866 4			B	
1975 4	$3/2^+, 5/2^+^\ddagger$		B	
2734 7			B	
3245 5	$3/2^+, 5/2^+^\ddagger$		B	
4275? 9	$3/2^+, 5/2^+^\ddagger$		B	XREF: B(?) Level exists assuming that the 4275 γ feeds the ground state of ^{35}Al (2014St18).

 † From a least-squares fit to γ -ray energies. ‡ From $L(^{36}\text{Si}, ^{35}\text{Al}) = 2$ of parallel momentum distributions of ^{35}Al residues (2014St18). $\gamma(^{35}\text{Al})$

$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\dagger$	E_f	J_f^π	Comments
802		802 4	100	0	$(5/2)^+$	
1007		1006 6	100	0	$(5/2)^+$	E_γ : weighted average of 1003 4 from $(^{36}\text{Si}, ^{35}\text{Al}\gamma)$ and 1020 9 from Coulomb excitation.
1866		859 4	100 8	1007		
		1064 4	22 6	802		
1975	$3/2^+, 5/2^+$	968 4	59 4	1007		
		1174 5	37 4	802		
		1972 6	100 7	0	$(5/2)^+$	

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Adopted Levels, Gammas (continued) $\gamma(^{35}\text{Al})$ (continued)

$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\dagger$	E_f	J_f^π	Comments
2734		1932 6	100	802		
3245	$3/2^+, 5/2^+$	2237 6	100 8	1007		
		2440 7	18.0 26	802		
		3250 8	42 5	0	$(5/2)^+$	
4275?	$3/2^+, 5/2^+$	4275 ‡ 9	100	0	$(5/2)^+$	2014St18 states that the 4275 γ transition could populate the ground state or one of the lowest two excited states. In the absence of any observed $\gamma\gamma$ coincidences, they tentatively place the 4275-keV γ transition feeding the ground state.

† From $^9\text{Be}(^{36}\text{Si}, ^{35}\text{Al}\gamma)$, unless otherwise noted.

‡ Placement of transition in the level scheme is uncertain.

Adopted Levels, Gammas

Legend

Level Scheme

Intensities: Relative photon branching from each level

 - - - - - $\rightarrow$ γ Decay (Uncertain)
